



MILL TEST CERTIFICATE

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100957570.0.2

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Date: April 16, 2019

MAVERICK TUBE CORPORATION
Hickman Plant
5020 N COUNTY RD 967
BLYTHEVILLE, AR 72315
(870) 776-5000 tel
(870) 776-5008 fax

Customer: Edgen Murray % Midwest, Kennedy Ave, SCHERERVILLE, IN 46375		Customer's Order Item: 31237105/SC/20		Customer's Reference: 241652		Manufacturer's Works Order N°: N/A	
Manufacturing Process: HFW		Product Type: Line Pipe		Surface: Fusion - Bond			
Standard or Specification: API 5L 45TH EDITION				Steel Grade: X52M PSL 2		Ends: Plain End Bevelled	
Dimensions: 10.750 x 0.500 in		Schedule: N/A		Length: Double Random		Quantity: 18 PCS 740 FT 40583 LB	
						Nominal Weight: 54.79 LB/FT	

TENSILE TEST

Heat N°	Sample N°	Zone	Lot N°	Specimen condition				Specimen dimensions	Test temp	Y.S.		U.T.S.	Elongation			
				Ls	Sc	Type	Ori			Size	F	.5% EUL	Req.	Lo	Min.	Obt.
												Min: 52 200	Min: 66 700			
												Max: 76 900	Max: 110 200			
									psi			in	%	%		
1191546	141549	E1	1191546	B	AM	Ss	T	1.500	RT	63 004	68 436	2.0	29.0	41.5		
1191547	141550	E1	1191547	B	AM	Ss	T	1.500	RT	64 897	69 449	2.0	29.0	40.1		
1191548	141551	E1	1191548	B	AM	Ss	T	1.500	RT	63 204	67 641	2.0	29.0	39.7		

AM: As manufactured	Ls: Location of sample	Ori: Orientation	Ss: Strip specimen
B: Body	Max: Maximum	Req: Required	T: Transversal
E1 / E2: Ends of Sampling	Min: Minimum	RT: Room temperature	U.T.S: Ultimate tensile strength
Lo: Initial length	Obt: Obtained	Sc: Specimen condition	Y.S: Yield strength

Heat N°	Sample N°	Zone	Lot N°	Specimen condition				Specimen dimensions	Test temp	U.T.S.
				Ls	Sc	Type	Ori			Req.
										Min 66 700
								Size		Max 110 200
								in	F	psi
1191546	141549	E1	1191546	W	AM	Ss	T	1.500	RT	73 427
1191547	141550	E1	1191547	W	AM	Ss	T	1.500	RT	76 203
1191548	141551	E1	1191548	W	AM	Ss	T	1.500	RT	72 985

AM: As manufactured	Min: Minimum	Sc: Specimen condition	W: Weld
E1 / E2: Ends of Sampling	Ori: Orientation	Ss: Strip specimen	

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Customer: Edgen Murray % Midwest, Kennedy Ave, SCHERERVILLE, IN 46375		Customer's Order Item: 31237105/SO/20		Customer's Reference: 241652		Manufacturer's Works Order N°: N/A	
Manufacturing Process: HFW		Product Type: Line Pipe		Surface: Fusion - Bond		Ends: Plain End Bevelled	
Standard or Specification: API 5L 45TH EDITION		Steel Grade: X52M PSL 2		Quantity: 18 PCS 740 FT		Nominal Weight: 54.79 LB/FT	
Dimensions: 10.750 x 0.500 in		Schedule: N/A		Length: Double Random		40583 LB	

TENSILE TEST

Ls: Location of sample Req: Required T: Transversal
Max: Maximum RT: Room temperature U.T.S: Ultimate tensile strength

CHEMICAL COMPOSITION

				Composition %																		
				X 1																		
				C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Al	Sn	Nb	V	Ti	Ca	B	N	CE-Pcm	
Heat N°	Sample N°	Lot N°	LS	H	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
				0.22	1.4	0.025	0.015	0.45	0.5	0.3	0.3	0.15	—	—	—	—	—	—	—	0.001	—	0.25
				P	Max	Min	0.22	1.4	0.025	0.015	0.45	0.5	0.3	0.3	0.15	—	—	—	—	0.001	—	0.25
					Min		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
1191546	3972402	1191546	B.M.	H	0.04	1.07	0.009	0.001	0.21	0.1	0.03	0.04	0.01	0.03	0	0.033	0.004	0.014	0.001	0	0.007	0.109
1191547	3972403	1191547	B.M.	H	0.04	1.07	0.01	0.001	0.17	0.1	0.03	0.04	0.01	0.03	0	0.036	0.003	0.012	0.001	0	0.006	0.108
1191548	3972401	1191548	B.M.	H	0.04	1.07	0.008	0.001	0.2	0.09	0.03	0.03	0.01	0.03	0	0.035	0.003	0.013	0.001	0	0.008	0.108
1191546	141376	1191546	B.M.	P	0.037	1.19	0.004	0.002	0.19	0.09	0.04	0.04	0.01	0.041	0	0.032	0.003	0.01	0.002	0.001	0.005	0.116
1191546	141377	1191546	B.M.	P	0.039	1.19	0.004	0.002	0.19	0.09	0.04	0.04	0.01	0.037	0	0.032	0.003	0.01	0.002	0.001	0.004	0.118
1191547	141372	1191547	B.M.	P	0.034	1.2	0.004	0.002	0.17	0.09	0.04	0.04	0.01	0.036	0	0.037	0.003	0.009	0.002	0.001	0.002	0.113
1191547	141373	1191547	B.M.	P	0.036	1.19	0.004	0.002	0.18	0.09	0.04	0.04	0.01	0.035	0	0.035	0.003	0.009	0.002	0.001	0.005	0.115
1191548	141370	1191548	B.M.	P	0.031	1.19	0.003	0.001	0.2	0.08	0.03	0.03	0.01	0.042	0	0.036	0.003	0.01	0.002	0.001	0.009	0.109
1191548	141371	1191548	B.M.	P	0.039	1.18	0.004	0.002	0.18	0.09	0.04	0.04	0.01	0.036	0	0.036	0.003	0.009	0.002	0.001	0.011	0.117

B.M: Base metal

H: Heat

Max: Maximum

P: Product

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MAVERICK TUBE CORPORATION
Hickman Plant
5020 N COUNTY RD 567
BLYTHEVILLE, AR 72315
(870) 778-5000 tel
(870) 778-5008 fax

Customer: Edgen Murray % Midwest, Kennedy Ave, SCHERERVILLE, IN 46375			Customer's Order Item: 31237105/SO/20		Customer's Reference: 241652		Manufacturer's Works Order N°: N/A		
Manufacturing Process: HFW			Product Type: Line Pipe					Surface: Fusion - Bond	
Standard or Specification: API 5L 45TH EDITION					Steel Grade: X52M PSL 2		Ends: Plain End Bevelled		
Dimensions: 10.750 x 0.500 in		Schedule: N/A	Length: Double Random		Quantity: 18 PCS 740 FT 40583 LB		Nominal Weight: 54.79 LB/FT		

CHEMICAL COMPOSITION

CE-Pcm: CE-Pcm

Ls: Location of sample

Min: Minimum

THROUGH WALL HARDNESS

Required values			Individuals			Average			Hardness type:HRBW		
			Min: -	Max: 91.0	Var: --	Min: --	Max: --	Var: --			
Heat N°	Sample N°	Zone	Lot N°	Pipe N°	Specimen condition		Quad.	Scale	Avg.		
					Ls	Sc					
1191546	141549	-	1191546	-	B	AM	-	HRBW	84.7		
1191547	141550	-	1191547	-	B	AM	-	HRBW	86.2		
1191548	141551	-	1191548	-	B	AM	-	HRBW	86.0		

AM: As manufactured

Ls: Location of sample

Quad: Quadrant

Avg: Average

Max: Maximum

Sc: Specimen condition

B: Body

Min: Minimum

Var: Variation

IMPACT TEST

Type: Charpy V			Orientation: T				Unit: ft.lb				Shear area %			
Position:			Temp: 32		Unit F		Ind.Min.Reg: 11		Req.Min.Avg: 15		Ind.Min.Reg: -		Req.Min.Avg: -	
Heat N°	Sample N°	Lot N°	Ls	Sc	T.T F	Size	Results				Results			
							1	2	3	Avg.	1	2	3	Avg.
1191546	141549	1191546	B	AM	32	3/4	207	207	206	207	100	100	100	100
1191547	141550	1191547	B	AM	32	3/4	205	202	205	204	100	100	100	100

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MAVERICK TUBE CORPORATION
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5020 N COUNTY RD 967
BLYTHEVILLE, AR 72316
(870) 776-5000 tel
(870) 776-5008 fax

Customer: Edgen Murray % Midwest, Kennedy Ave, SCHERERVILLE, IN 46375			Customer's Order Item: 31237105/SO/20		Customer's Reference: 241652		Manufacturer's Works Order N°: N/A	
Manufacturing Process: HFW			Product Type: Line Pipe			Surface: Fusion - Bond		
Standard or Specification: API 5L 45TH EDITION					Steel Grade: X52M PSL 2		Ends: Plain End Bevelled	
Dimensions: 10.750 x 0.500 in		Schedule: N/A	Length: Double Random		Quantity: 18 PCS 740 FT 40583 LB		Nominal Weight: 54.79 LB/FT	

IMPACT TEST

Type: Charpy V			Orientation: T				Unit: ft.lb				Shear area %			
Position:			Temp: 32		Unit: F		Ind.Min.Req: 11		Req.Min.Avg: 15		Ind.Min.Req: -		Req.Min.Avg: -	
Heat N°	Sample N°	Lot N°	Ls	Sc	T.T	Size	Results				Results			
							1	2	3	Avg.	1	2	3	Avg.
1191548	141551	1191548	B	AM	32	3/4	204	202	204	203	100	100	100	100

AM: As manufactured

Avg: Average

B: Body

Ind. Min. Req: Individual Minimum Required

Ls: Location of sample

Req. Min. Avg: Required minimum average

Sc: Specimen condition

T.T: Test temperature

Temp: Temperature

FLATTENING TEST

Standard				Standard				Standard			
Heat N°	Lot N°	Sample N°	Result	Heat N°	Lot N°	Sample N°	Result	Heat N°	Lot N°	Sample N°	Result
1191546	1191546	141549	Good	1191547	1191547	141550	Good	1191548	1191548	141551	Good

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Customer: Edgen Murray % Midwest, Kennedy Ave, SCHERERVILLE, IN 46375		Customer's Order Item: 31237105/SO/20		Customer's Reference: 241652		Manufacturer's Works Order N°: N/A	
Manufacturing Process: HFW		Product Type: Line Pipe		Surface: fusion - Bond			
Standard or Specification: API 5L 45TH EDITION		Steel Grade: X52M PSL 2		Ends: Plain End Bevelled			
Dimensions: 10.750 x 0.500 in	Schedule: N/A	Length: Double Random	Quantity: 18 PCS 740 FT	40583 LB	Nominal Weight: 54.79 LB/FT		

HYDROSTATIC TEST

Pressure		Time	Results
Unit	Value	Seconds	
psi	2 970	5	Satisfactory

SUPPLEMENTARY INFORMATION

Supplementary Information
This material has been manufactured, sampled, tested and inspected in accordance with this specification and has been found to meet the requirements called for by the above order. This product is silicon and/or aluminum fully killed. NDT OK 1/1 with ID/OD N-10 Notches verified by 1/8" FDIH INSPECTION CERTIFICATION 3.1 IN ACCORDANCE WITH EN 10204-2004

Heat Treatment
The minimum weld seam heat treatment temperature is 1650F.

Steel Vendor Name and Manufactured
1191546, NUCOR STEEL HICKMAN, ARMOREL, AR, Melted and Manufactured in USA
1191547, NUCOR STEEL HICKMAN, ARMOREL, AR, Melted and Manufactured in USA

Supplementary Information
COMPLIANT TO ASTM A53-12 GRADE B AND ASME SA53-13 GRADE B Material meets the maximum hardness requirement of section A 2.1.2 of NACE MRO 175 BODY TENSILE SOURCED 180° FROM WELD LINE; CVN SAMPLES SOURCED 90° FROM WELD LINE, CVN SAMPLES TESTED BY 8mm STRIKER

Steel Vendor Name and Manufactured
1191548, NUCOR STEEL HICKMAN, ARMOREL, AR, Melted and Manufactured in USA

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Standard or Specification: API 5L 45TH EDITION				Steel Grade: X52M PSL 2		Ends: Plain End Bevelled	
Dimensions: 10.750 x 0.500 in		Schedule: N/A		Length: Double Random		Quantity: 18 PCS 740 FT 40583 LB	
						Nominal Weight: 54.79 LB/FT	

SUMMARY REPORT

Heat N°	Quantity	Length	Weight	Heat N°	Quantity	Length	Weight	Heat N°	Quantity	Length	Weight
	(Pieces)	FT	LB		(Pieces)	FT	LB		(Pieces)	FT	LB
1191546	1	43.419	2.379.594	1191547	1	43.428	2.380.135	1191548	16	653.652	35.823.919

This is to certify that the product described here has been manufactured, sampled, tested, and inspected in accordance with purchaser order requirements. This certificate is not a declaration of origin nor may it be used as a declaration of origin.

	CUSTOMER . THIRD PARTY	TENARIS QUALITY DEPARTMENT SIGNATURE	
	INSPECTION COMPANY COMPAÑIA DE INSPECCIÓN Company Name: N/A Employee Name: N/A	 QUALITY CERTIFICATION DEPT. MCCORMICK Billy	 CHIEF OF QUALITY CERTIFICATION DEPT. DOVER Michael

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